WESTMINSTER
International University in Tashkent

System Analysis & Design

Lecture 4

Agenda

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1. Last week recap
2. Requirements definition
3. Requirement types
4. Requirements gathering process
5. Recommended reading

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Recap

What did we discuss last week?

System Develop Methodologies

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Structural:

- Waterfall
- Structured Software Development Methodology (SSDM)

Iterative and Incremental:

- RAD
- Spiral
- eXtreme Programming
- Scrum
- Etc.

Prototyping:

- Prototyping

Prototyping

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1. Types and differences of prototyping methods?
2. What are the differences between structured methods and prototyping?
3. When to use it?

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Requirements

Types of system requirements.

What is a requirement?

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Something that the product **must do** or a **quality** that the product **must have**.

1. **Functional requirements** – are things the product ***must do***.
   Examples: Ability to create a new blog post, edit existing posts, allow users to login, etc.

2. **Non-functional requirements** – are the ***qualities*** the product ***must have***.
   Examples: Prevailing color must be our corporate color, website should be able to process 100 concurrent users, interface must be easy to use, etc.

3. **Constraints** – are global issues that shape the requirements.
   Examples: Must be available in TAS-IX only, product must comply with ISO XXXX standard, legal constraints, etc.

NOTE: Collected/documented requirements must be unambiguous and non-vague.

Requirement types

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1. Functional
2. Non-functional *(Constraints are non-functional requirements)*

Functional Requirements

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- Specifications of the product's **functionality**
- **Actions** the product must take – check, calculate, record, retrieve
- Derived from the fundamental purpose of the product
- **Not** a quality

Can you think of an example?

Non-functional requirements

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- **Look and Feel** – the spirit of the product's appearance
- **Usability** – the product's ease to use, and any special usability considerations
- **Performance** – the operating environment of the product, and what considerations must be made for this environment
- **Operational** – the operating environment of the product, and what considerations must be made for this environment

Can you think of an example?

Non-functional requirements

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- **Maintainability and Portability** – expected changes and the time allowed to make them
- **Security** – the security and confidentiality of the product
- **Cultural and Political** – special requirements that come about because of the people involved in the product's development and operation
- **Legal** – what laws and standards apply to the product.

Can you think of an example?



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Requirements Gathering Process

How to collect the requirements.

Requirements Gathering Process

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1. Project Blastoff
2. Trawling for Knowledge
3. Prototyping and scenario modelling
4. Writing them down
5. Quality Gateway
6. Reusing requirements
7. Taking Stock of the specification
8. Post Mortem
9. Tailoring your process

1. Project Blast-off

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- Is a short burst of activity that puts together all the pieces that are necessary to get your project off to flying start and to ensure that your project is viable and well founded.
- Determines the precise purpose that the product is to fulfil.

Deliverables:

- Purpose of the project
- The client
- The customer
- The stakeholders
- The users
- Names
- Constraints
- Relevant facts and assumptions
- The scope of the work
- The estimated cost
- The risks

2. Trawling for Knowledge

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1. Studying the work that is currently done in response to a business event.
2. Learning the desires and aspirations of the adjacent systems involved in the business event.
3. Determining the best response that the organization can make to the business event.
4. Determining the way that the product will contribute to the desired response.
5. Determining and describing the requirements for the product.

3. Prototyping & scenario modelling

- The purpose of the ***prototype*** is to simulate aspects of the product and to make it real enough to help users think of requirements that might otherwise be missed.
- ***Scenario modelling*** is used to illustrate a situation. Contains a number of scenes or episodes that tell a detailed story of a specific situation.

4. Writing down

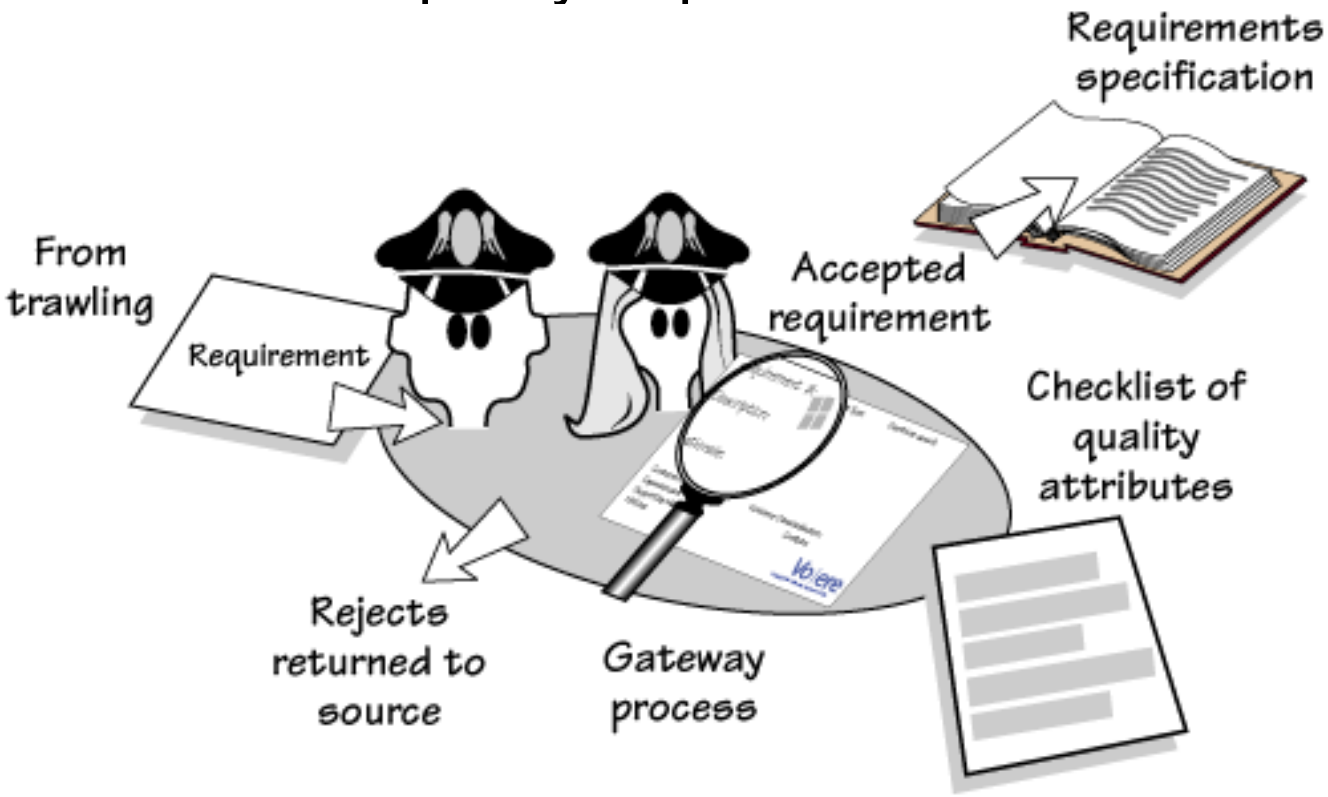
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- The process of documenting the requirements.
- Does not have to be “officially” documented. As far as it is clear and easy to understand.

5. Quality Gateway

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The process of filtering out non-feasible, incorrect or any requirement that does not fit your definition of quality requirement.



Robertson, J. and Robertson, S (2012). Mastering the Requirements Process. Addison-Wesley.

6. Reusing requirements

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- Entails making use of requirements that have been written for other projects.

- They can come from a number of sources:
  - a reuse library of specifications
  - other requirements specifications that are similar or in the same domain
  - informally from other people's experience.

7. Taking stock of the specification

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- Considers all of the requirements and their effect on each.

- The objective of the stock-take is to review the specification and find requirements that are:
  - Missing
  - Conflicting
  - Ambiguous

8. Post Mortem

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- Self-reflection and post mortem analysis.

- Questions you might want to ask:
  - What did we do right?
  - What did we do wrong?
  - If we had to do it again, what would you do differently?

9. Tailoring your process

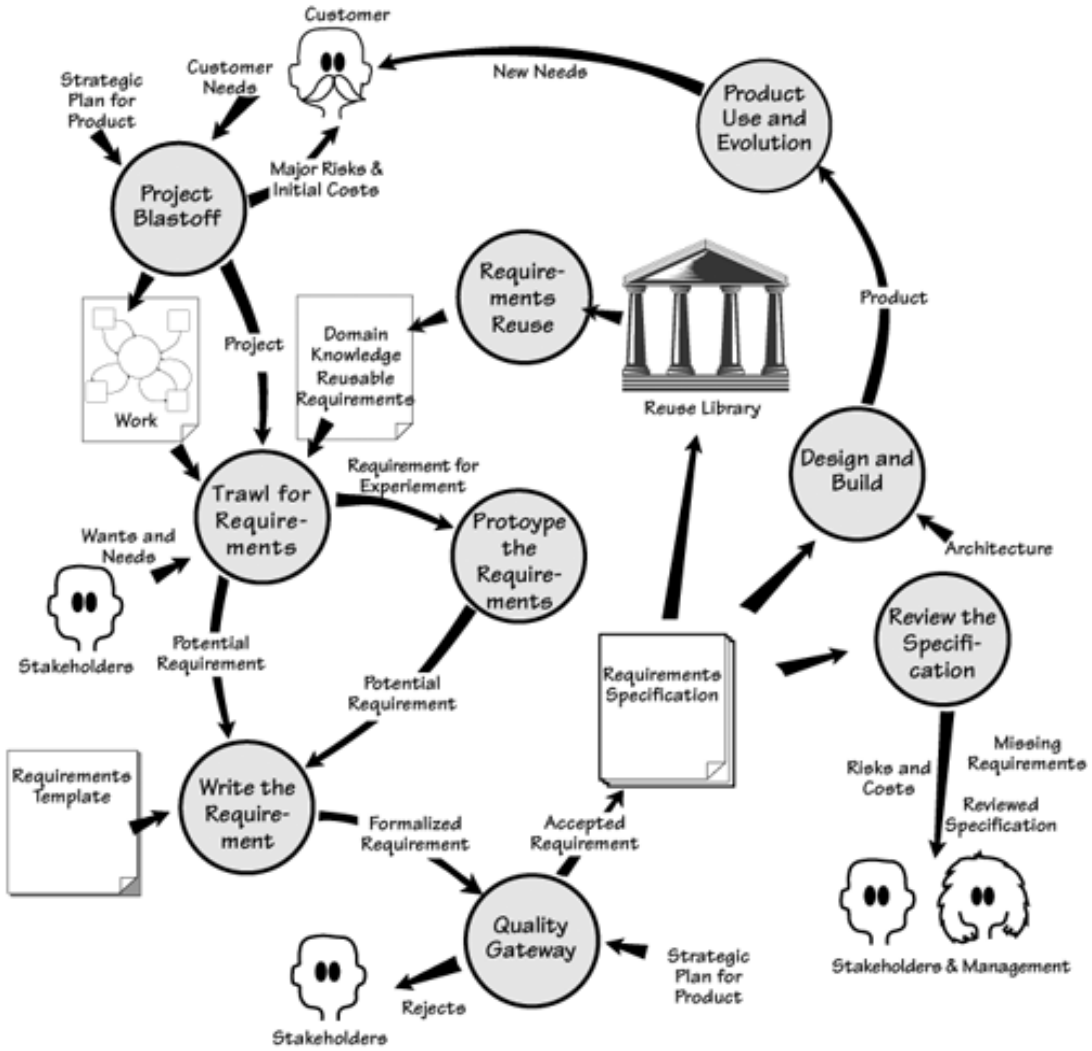
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- Once you understand the content and purpose of each deliverable, ask how each one would best be produced within your project environment using your sources.

- The tailoring task is determining how your project will produce them.

Visualizing Reqs Gathering

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Robertson, J. and Robertson, S (2012). Mastering the Requirements Process. Addison-Wesley.

Requirements Quality Check

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Can developer develop the feature just from reading my requirements and without contacting anyone else?



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Questions?

Recommended Reading

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- Robertson, S. and Robertson, J. (2013). *Mastering the requirements process*. Upper Saddle River, NJ: Addison-Wesley.

- Wikipedia, (2015). *Requirements analysis*. [online] Available at: https://en.wikipedia.org/wiki/Requirements_analysis [Accessed 19 Oct. 2015].

- Wikipedia, (2015). *Software requirements specification*. [online] Available at: https://en.wikipedia.org/wiki/Software_requirements_specification [Accessed 19 Oct. 2015].

- *Google…*



The End